

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2025/2026

OCTOBER EXAMINATION

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS

FRIDAY, 3 OCTOBER 2025

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS**Question 1**

EasyShop is a small-scale online retail platform currently offering approximately 100 products. With plans for global expansion and a significantly larger inventory, scalability becomes essential. The software engineering team must therefore design a robust search feature and select an algorithm that ensures both current efficiency and future scalability.

- a) Identify and explain the **TWO (2)** main types of complexity that the engineering team should analyse when selecting a search algorithm. (8 marks)
- b) **Table 1-1** illustrates the growth rate of running time as a function of input size n for three search algorithms labelled A, B, and C.

Algorithm	Growth-rate function : $f(n)$
A	$4n^2 + n + 50$
B	$80n + 250$
C	$10 \log n + 340$

Table 1-1 : Running time growth rate function for Algorithm A, B and C

- (i) Present the growth rate functions of Algorithms A, B, and C using Big-O notation. Show your working steps. (9 marks)
- (ii) Based on your answer to Question 1 b) (i), recommend the most suitable algorithm (A, B, or C) for the EasyShop engineering team to select. Provide a clear justification for your recommendation. (3 marks)
- c) The space/time trade-off is a fundamental concept in algorithm design.
- (i) Define the principle of space/time trade-off in the context of algorithm development. (2 marks)
- (ii) Provide an example of a space/time trade-off in algorithm design. (3 marks)

[Total: 25 marks]

Question 2

A set is a data structure that maintains a collection of distinct elements, ensuring that no duplicates are permitted within the collection.

- a) State **THREE (3)** main operations of the Set data structure used with multiple values. (3 marks)
- b) Compare the differences between HashSet, LinkedHashSet, and TreeSet in terms of ordering and performance. (6 marks)

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS**Question 2 (Continued)**

c) Write a complete Java program that meets the following requirements:

- Create two HashSet objects.
- Add the following three items to the first HashSet: "pencil", "eraser", and "ruler".
- Add the following three items to the second HashSet: "marker pen", "pen", and "eraser".
- Determine the common elements between the first and second HashSet objects.
- Display each common item using a loop, and also show the total number of common elements.

(16 marks)

[Total: 25 marks]

Question 3

a) Distinguish between the following terms:

- | | |
|---|-----------|
| (i) Static and dynamic data structures | (4 marks) |
| (ii) Static and dynamic memory allocation | (4 marks) |
| (iii) Queue and Stack | (4 marks) |

b) Given the binary tree shown in **Figure 3-1**, provide the output for each of the following traversal operations:

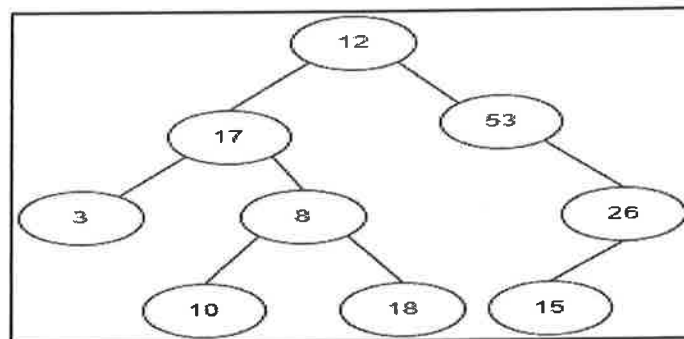


Figure 3-1 : binary tree

- | | |
|---|-------------|
| (i) In-order | (4.5 marks) |
| (ii) Post-order | (4.5 marks) |
| c) Discuss any TWO (2) advantages of using a linked list compared to an array. | (4 marks) |

[Total: 25 marks]

Question 4

A graph is a mathematical structure used to model relationships or associations among a set of entities, commonly representing real-world connections in various domains.

a) Using the graph provided in **Figure 4-1**, produce the corresponding :

(i) Adjacent matrix (5 marks)

(ii) Adjacent List (5 marks)

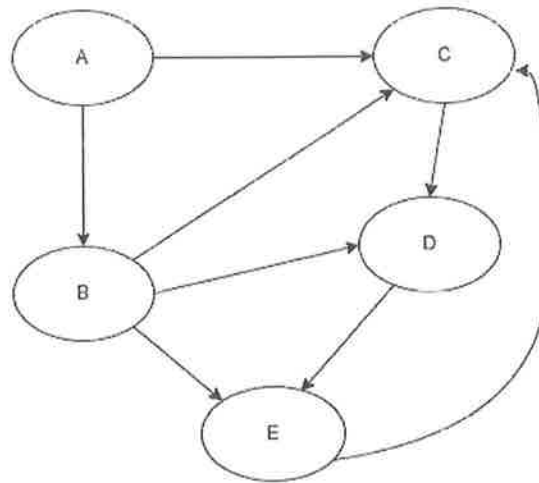


Figure 4-1 : Graph

b) With the help of diagrams, identify and illustrate any **THREE (3)** distinct types of graphs. (6 marks)

c) Provide any **THREE (3)** applications of graphs. (3 marks)

d) List and explain any **TWO (2)** commonly used graph traversal algorithms. (6 marks)

[Total: 25 marks]